Transformer Experiment Report

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1. Experimental Parameter Comparison

Stage	Parameters	Convergence Speed	Quality
Standard (Model1)	n_embd=128, n_layer=3	Fast	High noise, fragmented syntax, lacks logical flow.
Context Extension (Model2)	Increased Context Length	Moderate	Extended coherence, but severe hallucinations.
Increased Model Size (Model3)	n_embd=256, n_head=6, n_layer=6, dropout=0.1	Slower (Compute Intensive)	Emergence of dramatic structure and improved grammar.

2.Sample Output

1. Standard Model1

> **Scaled Model Generation:**
>
> To be, or not to be, that is the question:
> How callings. Who's weary weet upon thy borre;
> LAnd soul of him save him within; nor it words:

2. Context Extension Model2

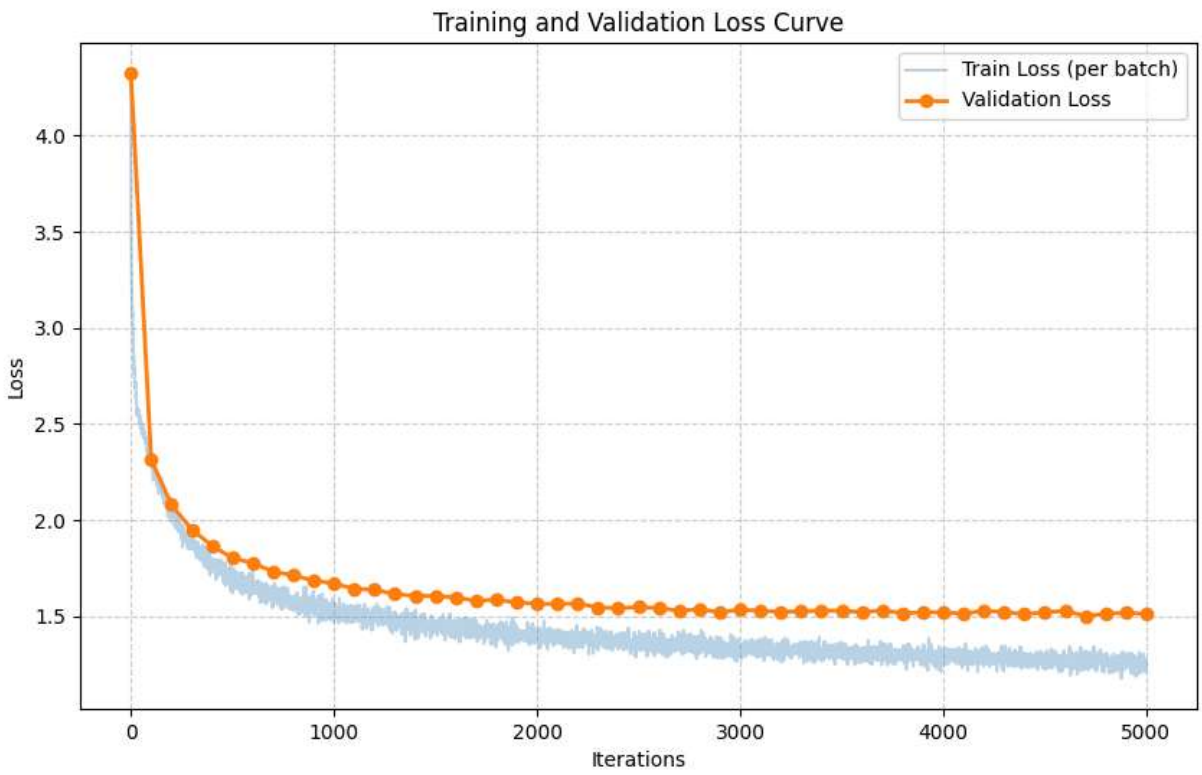
> **Scaled Model Generation:**

> To be, or not to be, that is the question:
> She's about unere some to serve thereof;
> When, you speak in drunking, call the cantinuates.

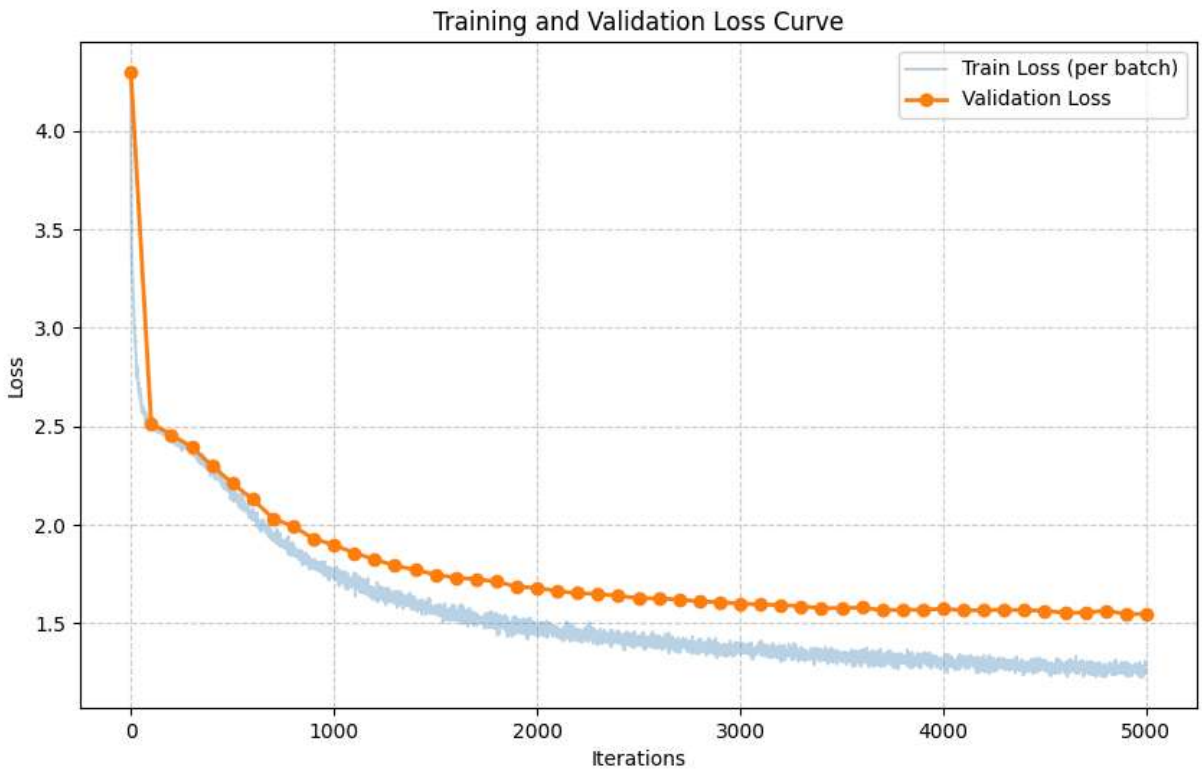
3. Increased Model Size Model3

> **Scaled Model Generation:**
>
> To be, or not to be, that is the question:
> 'It speak by thee ones; and why lay heard me thine.
> Lord CGREY: A good father, in my war; we are to do.

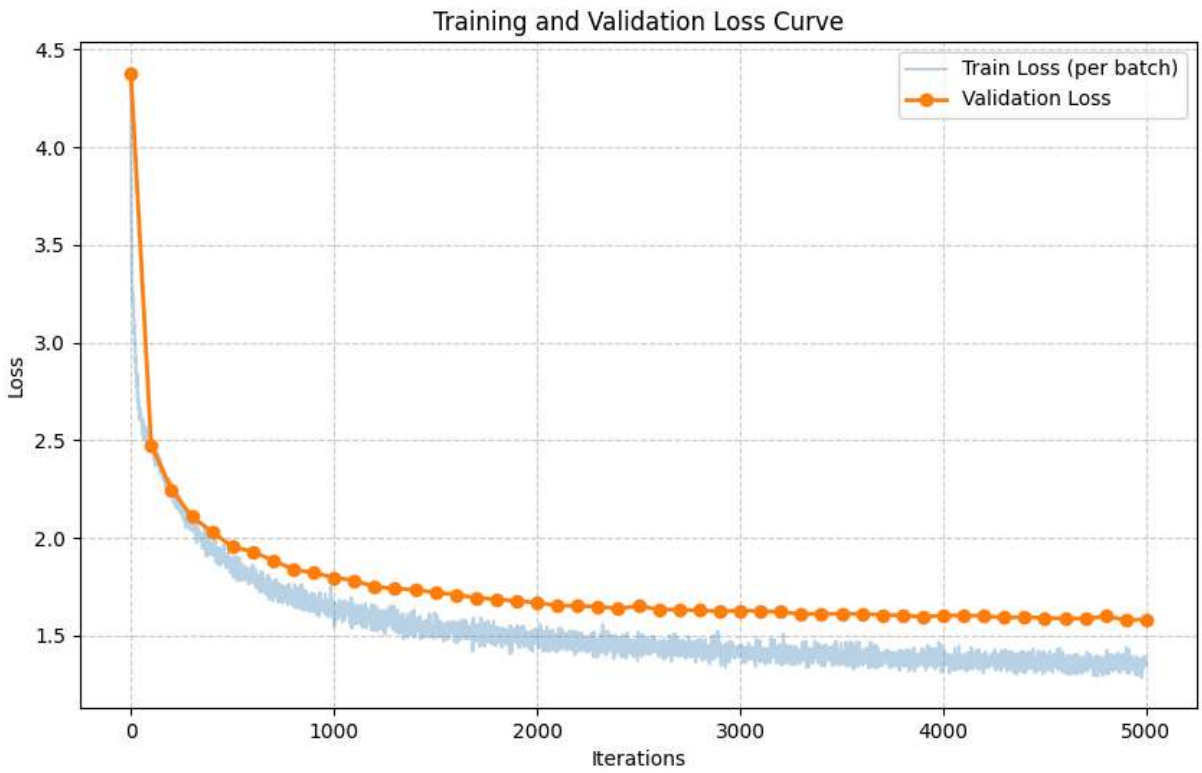
3. Training Loss Curve



Standard: Baseline performance.



Context: Improved coherence.



Capacity: Emergence of structure.

4. Conclusion

This report summarizes the experimental results of different transformer architectures and their impact on model performance. **Model Size (Capacity)** significantly improve model performance in understanding language and start to produce sentences with improved coherence and fluency with cost of training time and computation.